Evaluating the Impact of Sentiment Ambiguity on Static and Contextual Word Embeddings for Text Classification

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Abstract

Word embeddings, categorized as static (e.g., Word2Vec) or contextual (e.g., BERT), are fundamental for text classification tasks. This study tests the hypothesis that contextual embeddings outperform static embeddings in datasets with high ambiguity but offer comparable performance in low-ambiguity datasets. Using a simple Logistic Regression classifier, we evaluate embedding quality across datasets with varying ambiguity levels.

Our results reject this hypothesis: contextual embeddings outperform static embeddings only in low-ambiguity datasets, where sentiment markers are explicit. For medium- and high-ambiguity datasets, the difference in performance is not statistically significant, suggesting that the effectiveness of embeddings depends strongly on dataset complexity. These findings underscore the importance of aligning embedding selection with the specific characteristics of the task and dataset.

1 Introduction

Word embeddings provide dense vector representations of words, mapping them into a continuous space where relationships between words, such as semantic similarity, are encoded in their proximity. These embeddings serve as effective input features for a range of natural language processing (NLP) tasks, including *text classification* (Mikolov et al., 2013; Pennington et al., 2014), *machine translation* (Bahdanau et al., 2015), and *sequence learning* (Vaswani et al., 2017).

Broadly, word embeddings can be categorized into **context-independent embeddings** (e.g., Word2Vec (Mikolov et al., 2013), GloVe (Pennington et al., 2014)) and **context-dependent embeddings** (e.g., BERT (Devlin et al., 2019)). Static embeddings generate a single representation for each word, ignoring its contextual usage, while contextual embeddings dynamically adjust word representations based on surrounding context, enabling them to capture polysemy and nuanced meanings.

Ambiguity in Sentiment Classification

Ambiguity in sentiment classification refers to the degree to which the sentiment of a text is open to multiple valid interpretations, often due to linguistic phenomena such as polysemy, negation, idiomatic expressions, or lack of explicit sentiment markers. Ambiguous sentences may require additional linguistic or contextual information for accurate classification.

For instance:

- *"Not bad at all"* expresses a positive sentiment despite containing negative words.
- *"The bank was quiet today"* could refer to a financial institution or a riverside, requiring contextual disambiguation.

Ambiguity impacts the effectiveness of static embeddings, which provide a fixed representation for each word, as they cannot account for contextual variations. Contextual embeddings, on the other hand, dynamically adjust word representations based on the surrounding context, potentially resolving ambiguity by incorporating additional linguistic cues.

Hypothesis and Motivation

While contextual embeddings have demonstrated success in various NLP tasks, it is still useful to investigate how their performance compares to static embeddings across datasets with different linguistic characteristics. Exploring whether their ability to capture contextual information leads to measurable improvements over simpler static embeddings can provide valuable insights, especially for tasks like sentiment classification.

We hypothesize that **contextual embeddings, such as BERT, have a significant impact on**

datasets with high ambiguity, where contextual information is critical for disambiguating meaning, but offer limited advantage on low-ambiguity datasets where sentiment markers are explicit. To test this hypothesis, we evaluate both static and contextual embeddings on sentiment classification tasks across datasets with varying ambiguity levels.

Methodology Overview

To isolate the contribution of embeddings, we extract **sentence-level embeddings** as follows:

- For Word2Vec, sentence embeddings are generated by averaging word vectors across all tokens.
- For BERT, the embedding of the [CLS] token is used to represent the entire sentence.

The resulting embeddings are passed as input features to a **Logistic Regression classifier**, where we fine-tune the regularization parameter C. By treating contextual embeddings as **fixed feature extractors**, we minimize computational overhead while focusing on their ability to structure the embedding space for linear classification.

We evaluate this approach on sentiment classification datasets with varying ambiguity levels. Our study aims to provide insights into:

- The conditions under which contextual embeddings significantly outperform static embeddings.
- Whether the impact of contextual embeddings correlates with dataset ambiguity levels.
- When static embeddings remain competitive for simpler classification tasks.

The findings of this work contribute a resource-efficient framework for understanding and evaluating the impact of embedding types in text classification tasks.

2 Related Works

Several studies have compared static and contextual embeddings for sentiment classification tasks. Below, we summarize three key works and highlight how our work differs:

- Liu and Liu (2021): This study evaluates BERT and Word2Vec embeddings for sentiment classification using a Bidirectional Long Short-Term Memory (BLSTM) network, focusing on combining embeddings with deep neural architectures to improve performance. In contrast, our work uses a simple Logistic Regression classifier to isolate the intrinsic quality of embeddings, emphasizing their separability in the embedding space rather than the capacity of the downstream model.
- Konstantinov et al. (2021): This paper applies Word2Vec and BERT embeddings to analyze sentiment in social network texts, focusing on domain-specific datasets and a sentiment determination algorithm. Our work generalizes beyond domain-specific analysis, systematically comparing embedding performance across datasets with varying levels of ambiguity to identify broader patterns.
- Lee et al. (2022): This study compares TF-IDF, Word2Vec, and BERT embeddings for sentiment and subjectivity classification, highlighting BERT's strength in subjectivity tasks but showing competitive performance for simpler methods in sentiment classification. Our work focuses on dataset ambiguity as a key factor influencing embedding performance, providing a deeper analysis of when contextual embeddings offer significant advantages over static methods.

3 Methodology

To systematically compare static and contextual embeddings for sentiment classification, we designed an experiment that extracts sentence-level embeddings for each model and uses them as input features for a Logistic Regression classifier. From each sentence we've removed punctuation and converted everything to lowercase.

Embedding Extraction

We focus on two types of embeddings:

- **Word2Vec-`GoogleNews-vectors-negative300`**: Pre-trained Word2Vec embeddings are averaged across all tokens in a sentence to generate a single fixed-length vector representing the entire sentence (Mikolov et al., 2013).
- **BERT-`bert-base-uncased`**: The pre-trained BERT model is used to extract the embedding of the [CLS] token for each sentence, which

captures a context-aware representation of the sequence (Devlin et al., 2019).

Classification

The extracted sentence-level embeddings are passed as input features to a **Logistic Regression classifier**. To optimize the model, we perform a simple search over the regularization parameter C, fine-tuning it to maximize classification performance. This setup allows us to isolate the impact of the embeddings themselves, independent of the complexity of the downstream model.

Datasets

We evaluate the embeddings on three sentiment classification datasets with varying levels of ambiguity:

- **Amazon Reviews** (Zhang et al., 2015): A large-scale dataset of product reviews labeled as positive or negative, with clear sentiment markers (low ambiguity).
- **SST-2** (Socher et al., 2013): A fine-grained movie review dataset containing short sentences with nuanced linguistic structures (high ambiguity).
- **TweetEval Airlines Sentiment** (Barbieri et al., 2020): A Twitter-based dataset focusing on airline sentiment, with a mix of explicit and ambiguous sentiment markers (medium ambiguity).

Dataset	Train Size	Test Size	Ambiguity Level
Amazon Reviews	3,600,000	400,000	Low
SST-2	67,349	872	High
TweetEval Airlines Sentiment	9,764	1,000	Medium

Table 1: Summary of datasets used in this study

Due to computational resource constraints, we conducted our experiments on a randomly selected subset consisting of $\frac{1}{10}$ of the total size of each dataset. The samples were chosen randomly to ensure a representative distribution of the original data. While this reduced the overall dataset size, the random sampling approach preserves the essential characteristics of the datasets, such as ambiguity levels and sentiment distributions, allowing us to draw meaningful conclusions regarding the performance of contextual and static embeddings.

This diverse set of datasets allows us to study how embedding types perform under different linguistic complexities, ranging from explicit sentiment markers to nuanced and ambiguous expressions.

Evaluation Metrics and Statistical Significance

To evaluate the impact of fixed feature generation methods—*Word2Vec* (context-independent) and *BERT* (context-dependent)—on sentiment classification, we employ two key metrics: **accuracy** and **macro-F1 score**. Accuracy measures the overall correctness of predictions, while macro-F1 provides a balanced assessment of precision and recall across all classes, ensuring that class imbalances do not bias the evaluation. To determine whether the performance differences between Word2Vec and BERT are statistically significant, we calculate the **Wilson score interval** for both accuracy and macro-F1 at a **95% confidence level**. The Wilson interval provides a robust confidence margin for proportions, addressing the shortcomings of simpler approximations, particularly when sample sizes are finite (Wilson, 1927) (how we found this test is credited to (Brownlee, 2023)).

The confidence interval for a given metric $\hat{c}$ is calculated as:

$$c = \hat{c} \pm z\sqrt{\frac{\hat{c}(1-\hat{c})}{n} + \frac{z^2}{4n^2}}, \tag{1}$$

where n is the number of observations evaluated (equal to the number of test samples in our case), z is the constant critical value (1.96 for a 95% confidence level), and $\hat{c}$ is the observed metric value (e.g., accuracy or macro-F1).

By constructing 95% confidence intervals for the evaluation metrics, we can assess whether the observed performance differences are likely due to chance. If the confidence intervals for BERT and Word2Vec *do not overlap*, this indicates that the improvement is statistically significant. Such an analysis helps validate the impact of contextual embeddings on datasets with varying levels of ambiguity.

As a **baseline**, we implement a text classification model using **TF-IDF** features with Logistic Regression. The TF-IDF vectorizer converts the text into sparse numerical representations by assigning weights to words based on their frequency within and across documents.The Logistic Regression model is trained on these features using the liblinear solver, with hyperparameter tuning for regularization C through a simple search. This approach provides a simple, interpretable baseline to compare against more advanced word embedding methods like Word2Vec and BERT.

4 Results

The results of our experiments comparing BERT, Word2Vec, and the TF-IDF baseline for sentiment classification across three datasets are summarized in Table 2. From the evaluation metrics (accuracy and macro-F1 scores) and their corresponding Wilson score intervals, we make the following observations:

- **Amazon Reviews** (Low Ambiguity): BERT significantly outperforms Word2Vec, as evidenced by the **non-overlapping Wilson intervals** for both accuracy and macro-F1 scores. This indicates that contextual embeddings provide a clear advantage even in datasets dominated by explicit sentiment markers and minimal ambiguity.

- **SST-2** (High Ambiguity): While BERT achieves slightly higher scores than Word2Vec, the **Wilson intervals overlap**, suggesting that the difference is not statistically significant. This may reflect the extreme linguistic complexity of highly ambiguous datasets, which poses challenges for both models.

- **TweetEval Airlines Sentiment** (Medium Ambiguity): Both BERT and Word2Vec perform comparably, with **overlapping Wilson intervals** for both metrics. This result suggests that, in datasets with moderate ambiguity, the added contextual information provided by BERT does not yield a significant performance improvement over static embeddings.

These results reveal that the relative performance of contextual embeddings (BERT) and traditional embeddings (Word2Vec) is influenced by the level of ambiguity in the dataset. Interestingly, BERT's advantage is most pronounced in **low-ambiguity datasets**, where explicit sentiment markers dominate, contrary to our initial expectations. In contrast, its performance gains diminish on datasets with medium or high ambiguity, suggesting that other factors, such as model simplicity or dataset complexity, may influence these outcomes.

5 Discussion and Conclusion

The goal of this study was to analyze the relative performance of contextual embeddings (BERT) and traditional static embeddings (Word2Vec) for sentiment classification tasks across datasets with varying levels of ambiguity. Our results provide several key insights into the conditions under which contextual embeddings offer an advantage.

5.1 Discussion

Our experiments reveal that the performance advantage of contextual embeddings is not as generalizable as initially hypothesized. Contrary to our expectations, BERT only outperforms Word2Vec on **low-ambiguity datasets** (e.g., Amazon Reviews), where sentiment markers are explicit and context plays a minimal role. This surprising result suggests that the flexibility of contextual embeddings allows BERT to encode fine-grained signals even when sentiment cues are straightforward, providing a slight edge over Word2Vec's static representations.

In contrast, on **medium-ambiguity datasets** (e.g., TweetEval Airlines Sentiment), BERT and Word2Vec perform comparably, with overlapping Wilson confidence intervals. While we expected BERT to excel in these scenarios by capturing subtle nuances, its contextual strength does not appear to translate into statistically significant improvements. One possible explanation is that, in moderately ambiguous data, simpler static embeddings are sufficient to encode the sentiment markers, leaving little room for BERT to showcase its contextual capabilities.

For **high-ambiguity datasets** (e.g., SST-2), the Wilson intervals for BERT and Word2Vec also overlap, indicating no significant difference. The extreme linguistic complexity of such datasets—characterized by polysemy, conflicting sentiment cues, and intricate syntax—likely challenges both models. Additionally, the use of a simple Logistic Regression classifier may limit BERT's ability to exploit its richer embeddings, as a linear model may not fully leverage the contextual information provided by transformers.

These findings **reject our initial hypothesis**, which posited that BERT would demonstrate a clear advantage in high- and medium-ambiguity settings due to its ability to capture nuanced context. Instead, the results suggest that contextual embeddings only provide measurable benefits on

Dataset	Method	Accuracy	Macro-F1 Score
Amazon Reviews	Baseline	0.7778 ± 0.1322	0.6786 ± 0.1586
	BERT	0.8800 ± 0.0283	0.8800 ± 0.0283
	Word2Vec	0.7850 ± 0.0374	0.7849 ± 0.0353
SST-2	Baseline	0.8175 ± 0.0348	0.8175 ± 0.0325
	BERT	0.8326 ± 0.0233	0.8325 ± 0.0223
	Word2Vec	0.8039 ± 0.0250	0.8031 ± 0.0247
TweetEval Airlines Sentiment	Baseline	0.5065 ± 0.0279	0.4233 ± 0.0272
	BERT	0.6173 ± 0.0268	0.6072 ± 0.0264
	Word2Vec	0.6050 ± 0.0270	0.6016 ± 0.0264

Table 2: Comparison of Accuracy and Macro-F1 Scores (with Wilson intervals at a 95% confidence level) across Amazon Reviews, SST-2, and TweetEval Airlines Sentiment datasets.

low-ambiguity datasets, where they enhance even explicit signals. Future work could investigate whether this trend holds when more complex classifiers or alternative ambiguity metrics are applied. Furthermore, analyzing tailored datasets designed to isolate specific types of ambiguity could help clarify the role of contextual embeddings in challenging linguistic settings.

5.2 Limitations

There are several limitations to this work:

- **Dataset Scope**: We evaluated only three datasets, each representing specific levels of ambiguity as qualitatively assessed. However, the lack of a solid, quantitative metric for ambiguity means our categorization of datasets might not fully capture the nuances of linguistic complexity. This limitation suggests that our test may not be the most precise method to accurately characterize the impact of contextual embeddings. A more rigorous metric for ambiguity or tailored datasets designed to exhibit varying degrees of ambiguity explicitly could provide more definitive insights into the relationship between embedding types and their effectiveness.
- **Model Simplicity**: Logistic Regression was used as the downstream classifier to evaluate the linear separability of embeddings. While effective for isolating embedding quality, it may not fully capture non-linear relationships or complex clustering patterns. Incorporating non-linear models or clustering metrics could provide a more comprehensive analysis of embedding effectiveness.
- **Feature Extraction**: Contextual embeddings (BERT) were used as fixed feature extractors, without fine-tuning. While this approach isolates the embedding quality, fine-tuning could improve contextual embeddings' performance, particularly in high-ambiguity datasets.

5.3 Future Work

To extend this study, future work could:

- Explore additional datasets encompassing diverse linguistic phenomena, such as sarcasm, irony, or domain-specific texts.
- Investigate the impact of fine-tuning contextual embeddings on specific datasets to determine if this amplifies their performance advantage.
- Compare embeddings using more complex classifiers, such as neural networks, to understand how embedding quality interacts with downstream model complexity.

5.4 Conclusion

This work shows that the performance of contextual embeddings (BERT) and static embeddings (Word2Vec) depends on the level of ambiguity within the dataset. Surprisingly, contextual embeddings outperform static embeddings only in low-ambiguity datasets, where explicit sentiment markers dominate. In medium- and high-ambiguity settings, both embedding types perform comparably, highlighting the need to carefully consider dataset characteristics when selecting embeddings for sentiment classification tasks.

6 Contribution

Both members contributed equally to this project, working together on the coding, analysis, and report writing.

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